Ananya Joshi

ananya.joshi@example.com | +91 99480 24439 | https://github.com/ananyajoshi13 | https://linkedin.com/in/ananyajoshi13 | https://ananyajoshi13.dev

EDUCATION

Master of Technology in Software Systems

Indian Institute of Technology Madras
CGPA: 9.00/10.0 (2022-2026)

SKILLS

Languages
Go, Java, Python, TypeScript, C++, SQL

Frameworks & Systems
Spring Boot, Node.js, FastAPI, React, Next.js, gRPC

Databases & Storage
PostgreSQL, Redis, MongoDB, Apache Kafka, Elasticsearch

Cloud & DevOps
Docker, Kubernetes, AWS (EC2, S3, RDS), Git, CI/CD Pipelines

EXPERIENCE

Full Stack Software Engineer — Razorpay Software Pvt. Ltd. *Jun 2025 – Present*

- Optimized database queries and added composite b-tree indexes on partitioned ledger tables, lowering average query execution time from 420ms to 35ms.
- Implemented automated circuit-breaker pattern with Resilience4j across 8 downstream microservices, preventing cascade outages during third-party gateway downtime.
- Migrated monolithic checkout service to gRPC microservices, shrinking cloud compute costs by 32% and reducing deployment cycle times from 4 hours to 12 minutes.

PROJECTS

Peer-to-Peer File Transfer Protocol with NAT Traversal [Rust, STUN/TURN, UDP, Tokio, Cryptography]

- Developed user-space P2P file synchronization tool implementing UDP hole-punching and ICE protocol to bypass restrictive symmetric NATs.
- Utilized ChaCha20-Poly1305 authenticated encryption to guarantee confidentiality and packet integrity over public Wi-Fi networks.
- Achieved wire-speed throughput of 940 Mbps over local gigabit connections using zero-copy memory-mapped file buffers.

High-Concurrency Real-Time Collaboration Canvas [TypeScript, WebSocket, CRDT, React, Node.js]

- Built multi-user collaborative workspace using Conflict-free Replicated Data Types (CRDTs) to ensure convergence across disconnected edits.
- Engineered WebSocket server in Node.js with delta-compression protocol, reducing network bandwidth consumption by 64% during intense cursor movements.
- Authored comprehensive property-based test suite with fast-check verifying eventual consistency across simulated random network partitions.